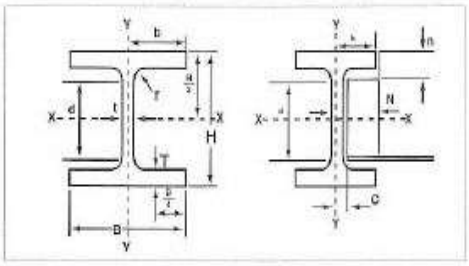
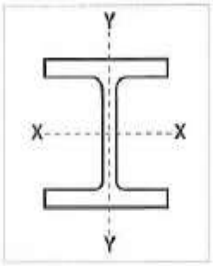
Dimensions and Properties



4a (i) Beams (Imperial Units) - Part I

	Designation Size		Unit Weight		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
							Flange	Web				Flange	Web	End Clearance	Notch	
					H	B	T	t	r	d	A	b/T	d/t	C	N	n
	In	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W5	5 x 3	(127 x 76)	9	13.4	127.0	76.0	7.6	4.0	7.6	96.6	16.5	5.00	24.2	4	46	15
W6	6 x 3	(152 x 76)	9	13.4	150.0	75.0	7.0	5.0	8.0	120.0	17.8	5.36	24.0	5	45	15
			12	17.9	155.0	75.0	8.5	6.0	8.0	122.0	21.6	4.41	20.3	5	45	17
	6 x 3 1/2	(152 x 89)	11	16.4	152.4	88.7	7.7	4.5	7.6	121.8	20.3	5.76	27.1	4	52	15
	6 x 4	(152 x 102)	9	13.4	149.9	100.1	5.5	4.3	6.4	126.3	17.3	9.17	29.2	4	58	12
			12	17.9	153.2	101.6	7.1	5.8	6.4	126.3	22.9	7.14	21.6	5	58	13
			16	23.8	159.5	102.4	10.2	6.6	6.4	126.4	30.4	5.02	19.2	5	58	17
W7	7 x 3 1/2	(178 x 89)	11	16.4	173.0	90.0	7.0	4.5	8.9	141.2	20.4	6.43	31.4	4	53	16
			12	17.9	175.0	90.0	8.0	5.0	8.9	141.2	23.0	5.63	28.2	5	53	17
			15	22.3	179.0	90.0	10.0	6.0	8.9	141.2	28.2	4.50	23.5	5	52	19
	7 x 4	(178 x 102)	13	19.3	177.8	101.2	7.9	4.8	7.6	146.8	24.3	6.41	30.6	4	58	16
W8	8 x 4	(203 x 102)	10	14.9	200.4	100.1	5.2	4.3	7.6	174.7	19.1	9.61	40.4	4	58	13
			13	19.3	202.9	101.6	6.5	5.8	7.6	174.7	24.8	7.84	29.9	5	58	14
			15	22.3	206.0	102.0	8.0	6.2	7.6	174.8	28.6	6.38	28.1	5	58	16
			16	23.1	203.2	101.8	9.3	5.4	7.6	169.4	29.4	5.47	31.4	5	58	17
	8 x 5 1/4	(203 x 133)	14	20.8	203.0	133.0	6.4	5.0	8.0	174.2	27.1	10.39	34.8	5	74	14
			15	22.3	202.0	133.0	7.0	5.0	8.9	170.2	28.7	9.50	34.0	5	74	16
			17	25.1	203.2	133.2	7.8	5.7	7.6	172.4	32.0	8.54	30.2	5	74	15
			18	26.8	206.8	133.3	8.4	5.8	7.6	174.8	33.9	7.95	29.9	5	74	16
			20	29.8	206.8	133.9	9.6	6.4	7.6	172.4	38.2	6.97	26.9	5	74	17
			21	31.3	210.3	133.9	10.1	6.4	7.6	174.9	39.6	6.63	27.5	5	74	18
	8 x 6 1/2	(203 x 165)	24	35.7	201.4	165.0	10.1	6.2	10.2	160.8	45.5	8.17	25.9	5	89	20
			28	41.7	204.7	166.0	11.8	7.2	10.2	160.7	53.2	7.03	22.2	6	89	22
W10	10 x 4	(254 x 102)	12	17.9	250.7	100.6	5.3	4.8	7.6	224.8	22.8	9.44	46.5	4	58	13
			15	22.3	254.0	101.6	6.8	5.7	7.6	225.2	28.0	7.47	39.5	5	58	14
			17	25.3	257.2	101.9	8.4	6.0	7.6	225.2	32.0	6.07	37.5	5	58	16
			19	28.3	260.4	102.2	10.0	6.3	7.6	225.2	36.1	5.11	35.7	5	58	18
	10 x 5 3/4	(254 x 146)	16	23.8	253.0	145.0	6.4	5.0	7.6	224.2	31.1	11.33	44.8	5	80	14
			21	31.3	251.4	146.1	8.6	6.0	7.6	219.0	39.7	8.49	36.5	5	80	16
			22	32.7	258.3	146.0	9.1	6.1	7.6	224.8	41.8	7.99	36.8	5	80	17
			25	37.2	256.0	146.4	10.9	6.3	7.6	219.0	47.2	6.72	34.8	5	80	19
			26	38.7	262.4	146.6	11.1	6.6	7.6	225.0	48.9	6.60	34.1	5	80	19
			29	43.2	259.6	147.3	12.7	7.2	7.6	219.0	54.8	5.80	30.4	6	80	20
			30	44.6	265.9	147.6	12.9	7.6	7.6	224.9	56.9	5.72	29.5	6	80	21
	10 x 8	(254 x 203)	33	49.1	247.1	202.2	11.0	7.4	12.7	199.7	62.5	9.19	27.1	6	107	24
			39	58.0	252.0	202.8	13.4	8.0	12.7	199.8	73.8	7.57	25.0	6	107	26
			42	62.5	249.4	207.0	12.2	12.2	12.7	199.6	79.3	8.49	16.4	8	107	25
			45	67.0	256.5	203.7	15.7	8.9	12.7	199.7	85.4	6.49	22.5	6	107	28
			57	84.8	258.1	211.3	16.5	16.5	12.7	199.7	108	6.40	12.1	10	107	29
W12	12 x 4	(305 x 102)	14	20.8	302.5	100.8	5.7	5.1	7.6	275.8	26.8	8.83	54.3	5	58	13
			16	23.8	304.5	101.3	6.7	5.6	7.6	275.8	30.4	7.53	49.3	5	58	14
			16.5	24.6	305.1	101.6	7.0	5.8	7.6	275.9	31.6	7.26	47.6	5	58	15
			19	28.3	308.7	101.8	8.8	6.0	7.6	275.9	35.9	5.78	46.0	5	58	16
			22	32.7	312.7	102.4	10.8	6.6	7.6	275.9	41.8	4.74	41.8	5	58	18
	12 x 5	(305 x 127)	25	37.2	304.4	123.3	10.7	7.1	8.9	265.2	47.2	5.76	37.4	6	68	20
			28	41.7	307.2	124.3	12.1	8.0	8.9	265.2	53.4	5.14	33.2	6	68	21
			32	47.6	311.0	125.3	14.0	9.0	8.9	265.2	61.2	4.48	29.5	7	68	23

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
*Please contact us for further clarification.



4a (i) Beams (Imperial Units) - Part II

Unit Weight		Surface Area per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation Size	
			Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J		
lb/ft	kg/m	m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			dm^6	cm^4	in	
9	13.39	0.537	473	55.8	5.35	1.84	74.6	14.7	84.2	22.6	0.895	16.3	0.00199	2.85	5 x 3	W5
9	13.39	0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.00253	2.81	6 x 3	W6
12	17.86	0.584	841	60.2	6.24	1.67	109	16.0	126	25.4	0.875	17.6	0.00323	4.81		
11	16.37	0.638	834	89.8	6.41	2.10	109	20.2	123	31.2	0.890	19.6	0.00470	3.56	6 x 3 1/2	
9	13.39	0.681	683	91.4	6.29	2.30	91.1	18.3	102	28.1	0.873	26.2	0.00477	1.69	6 x 4	
12	17.86	0.690	918	125	6.33	2.33	120	24.5	136	38.0	0.869	20.4	0.00665	3.76		
16	23.81	0.704	1330	183	6.61	2.45	167	35.7	190	55.2	0.882	15.4	0.0102	9.12		
11	16.37	0.682	1060	85.4	7.20	2.04	123	19.0	138	29.4	0.884	23.9	0.00588	3.15	7 x 3 1/2	W7
12	17.86	0.685	1213	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.00680	4.48		
15	22.32	0.691	1529	122	7.36	2.08	171	27.1	195	42.3	0.886	17.8	0.00871	8.16		
13	19.35	0.738	1356	137	7.48	2.37	153	27.0	171	41.6	0.888	22.6	0.00987	4.41	7 x 4	
10	14.88	0.779	1284	87.3	8.19	2.14	128	17.4	145	27.2	0.867	36.3	0.00832	1.78	8 x 4	W8
13	19.35	0.787	1647	114	8.16	2.14	162	22.4	187	35.3	0.861	29.0	0.0110	3.63		
15	22.32	0.794	1999	142	8.36	2.23	194	27.8	222	43.7	0.869	25.2	0.0139	5.68		
16	23.81	0.790	2105	164	8.46	2.36	207	32.2	234	49.8	0.888	22.5	0.0154	7.02		
14	20.83	0.914	1980	251	8.55	3.05	195	37.8	218	58.0	0.874	29.9	0.0243	3.74	8 x 5 1/4	
15	22.32	0.911	2105	275	8.56	3.09	208	41.3	232	63.4	0.879	27.9	0.0261	4.50		
17	25.30	0.915	2340	308	8.56	3.10	230	46.2	258	70.9	0.877	25.6	0.0294	5.96		
18	26.79	0.922	2578	331	8.72	3.12	249	49.7	279	76.3	0.880	24.5	0.0326	7.13		
20	29.76	0.923	2896	385	8.71	3.17	280	57.5	314	88.2	0.881	21.5	0.0374	10.3		
21	31.25	0.930	3119	405	8.87	3.20	297	60.4	333	92.7	0.885	21.0	0.0405	11.6		
24	35.72	1.03	3430	757	8.68	4.08	341	91.7	378	140	0.875	19.3	0.0693	14.3	8 x 6 1/2	
28	41.67	1.04	4077	901	8.76	4.12	398	109	445	165	0.875	16.8	0.0838	22.4		
12	17.86	0.881	2241	90.8	9.91	1.99	179	18.0	207	28.6	0.851	44.0	0.0137	2.28	10 x 4	W10
15	22.32	0.890	2841	119	10.1	2.06	224	23.5	259	37.3	0.856	36.4	0.0182	4.15		
17	25.30	0.897	3415	149	10.3	2.15	266	29.2	306	46.0	0.866	31.5	0.0230	6.42		
19	28.28	0.904	4005	179	10.5	2.22	308	34.9	353	54.8	0.874	27.5	0.0280	9.57		
16	23.81	1.06	3477	326	10.6	3.23	275	44.9	308	69.0	0.874	38.2	0.0494	4.16	10 x 5 3/4	
21	31.25	1.06	4413	448	10.5	3.36	351	61.3	393	94.1	0.880	29.6	0.0660	8.55		
22	32.74	1.08	4917	475	10.8	3.37	381	65.0	426	100	0.882	28.9	0.0737	9.96		
25	37.20	1.07	5537	571	10.8	3.48	433	78.0	483	119	0.890	24.3	0.0857	15.3		
26	38.69	1.08	5974	584	11.1	3.45	455	79.6	510	122	0.888	24.5	0.0921	16.5		
29	43.16	1.08	6544	677	10.9	3.52	504	92.0	566	141	0.891	21.2	0.103	23.9		
30	44.64	1.09	7048	692	11.1	3.49	530	93.8	597	144	0.888	21.3	0.111	25.6		
33	49.11	1.27	7071	1517	10.6	4.93	572	150	634	229	0.873	21.5	0.211	24.0	10 x 8	
39	58.04	1.28	8672	1865	10.8	5.03	688	184	765	280	0.878	18.3	0.265	40.2		
42	62.50	1.25	8430	1807	10.3	4.77	676	175	768	271	0.851	17.9	0.254	44.5		
45	66.97	1.29	10300	2214	11.0	5.09	803	217	898	331	0.880	15.9	0.321	62.3		
57	84.83	1.31	11930	2606	10.5	4.91	925	247	1067	385	0.849	13.6	0.380	109		
14	20.83	0.985	3682	97.9	11.7	1.91	243	19.4	286	31.1	0.842	50.8	0.0216	2.93	12 x 4	W12
16	23.81	0.990	4274	117	11.9	1.96	281	23.1	329	37.0	0.846	45.0	0.0260	4.27		
16.5	24.55	0.992	4455	123	11.9	1.97	292	24.2	342	38.8	0.846	43.4	0.0273	4.77		
19	28.28	1.00	5366	155	12.2	2.08	348	30.5	403	48.5	0.859	37.4	0.0349	7.40		
22	32.74	1.01	6501	194	12.5	2.15	416	37.9	481	60.0	0.866	31.6	0.0442	12.2		
25	37.20	1.07	7166	335	12.3	2.67	471	54.4	539	85.3	0.872	29.7	0.0723	14.8	12 x 5	
28	41.67	1.08	8196	389	12.4	2.70	534	62.6	614	98.4	0.872	26.5	0.0847	21.1		
32	47.62	1.09	9575	461	12.5	2.74	616	73.6	711	116	0.873	23.3	0.102	31.8		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.
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